MILITARY SPECIFICATION

PROJECTILE, GUIDED, 155MM, XM712
THERMAL BATTERY FOR

This amendment forms a part of Military Specification MIL-P-63235(AR), dated 31 August 1979, and is approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title: Add "M712" after the designation "XM712". Make this addition wherever it appears in the entire specification.

PAGE 2

* 2.1 DRAWINGS, Army: Delete "9301200 - Battery, Thermal..." and substitute with the following: "9304695 - Battery, Thermal...".

* 3.1 Third sentence: Delete it its entirety and substitute the following: "The +11VDC section is activated by two redundant inertial starter mechanisms and the ± 30 Vdc section are activated by an electrical initiator."

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3.2.1.1.2.1: Delete entire paragraph and substitute the following:

"3.2.1.1.2.1 +30 Volt source squib loads.- There shall be five 4.75 ± 0.68 ohm squib loads applied to the +30 volt source. The squib loads shall occur at 1, 2, 3, 4, and 5 seconds after initiation and shall have a duration of 20 ± 1 milliseconds. The +30 volt source output voltage shall be +28.0 volts minimum during applied squib loads and shall recover to within +0, -2 volts, of the same voltage level as that prior to the application of each of the respective squib loads, in a maximum of 75 milliseconds."

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3.2.1.1.4: Delete entire paragraph and substitute the following:

"3.2.1.1.4 Voltage variations (noise).- No voltage output source shall exhibit voltage variations, other than those due to load changes and source impedance changes during activation, except to the extent specified herein. The amplitude of any voltage variation shall not exceed six (6) volts peak to peak for the first five (5) seconds after the section has attained minimum specified voltage. From the first five (5) seconds to the end of specified life, voltage variations occurring more than once per second shall not exceed 0.5 volts peak to peak."

* 3.2.1.2.1 Delete in its entirety and substitute the following:

"3.2.1.2.1 +11 Volt source activation. The +11 volt source is activated by two redundant inertial starters. The inertial starters shall comply with the following requirements."

3.2.1.2.1.1: Delete entire paragraph and substitute the following:

"3.2.1.2.1.1 No-fire stimulus.- The inertial starter no-fire stimulus shall be 350 G minimum acceleration for a time period of 2 milliseconds, minimum (1/2 Sine pulse)."

3.2.1.2.1.2: Delete entire paragraph and substitute the following:

"3.2.1.2.1.2 All-fire stimulus.- The inertial starter all-fire stimulus shall be 900 $\pm$ 150 Gs acceleration for a time period of 15 milliseconds maximum (1/2 Sine pulse)."

3.2.1.2.2.2: Delete entire paragraph and substitute the following:

"3.2.1.2.2.2 No-Fire current.- The electrical initiator no-fire current shall be one (1) ampere dc minimum, for a minimum of 15 seconds at 69°F (21°C)."

3.2.1.2.2.3: Delete entire paragraph and substitute the following:

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"3.2.1.2.2.3 All-fire current.- The electrical initiator all-fire current range shall be 4.5 amperes dc maximum for 20±1 milliseconds."

3.2.1.4: Delete entire paragraph and substitute the following:

"3.2.1.4 Terminal strength.- Each battery terminal shall withstand a pull test without rupture or damage under an applied force of four plus or minus one pound (4±1)."

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3.2.2.6: Delete entire paragraph and substitute the following:

"3.2.2.6 Transportation vibration.- The thermal battery shall meet the requirements of 3.2.1 following exposure to sinusoidally cycled vibrations ranging from 2.5 Gs double amplitude to 5 Hz to 92 Hz, 0.0058 in. double amplitude at 93 Hz to 110 Hz and 3.5 Gs double amplitude a 111 to 200 Hz including sinusoidal dwell vibrations of ± 35 Gs at 110 Hz for 20 seconds in each of 3 axis."

* 3.3.1, third line: Delete "9301200" and substitute "9304695".

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* 4.5.2.1: Delete entire paragraph and substitute the following:

"4.5.2.1 Terminal strength.- Test terminal strength as specified in MIL-STD-202, Method 211, Test Condition A, using a force of four plus one pound minum zero pounds (4±1 lbs). After test, visually inspect terminals for rupture or damage and for compliance with the requirements of 3.2.1.4."

4.5.2.7: Delete entire paragraph and substitute the following:

"4.5.2.7 ±30 Volt source match no-fire.- Apply a direct current (dc) of 1.0 ampere minimum, across the match "M" terminals through the electrical initiator for 15 seconds, minimum, using a suitable power source. The battery shall meet the requirements of 3.2.1.2.2.2."

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4.5.2.8: Delete entire paragraph and substitute the following:

"4.5.2.8 +11 volt source no-fire.- The battery shall be mounted to an appropriate shock machine and the no-fire stimulus of 3.2.1.2.1.1 applied to the battery. The +11 volt section of the battery shall not be activated."

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* 4.5.2.9 1: Delete entire paragraph and substitute the following:

"4.5.2.9.1 Activation test.- The battery shall be mounted to an appropriate shock machine and the 11 volt section activated by applying the all-fire stimulus of 3.2.1.2.1.2. The ±30 volt section of the battery shall be activated by applying the 11 volt section output across the ±30 volt section electrical initiator, through a series resistor of 2.1 ohms at 145°F and 1.5 ohms at -25°F, at 200 milli-seconds after the 11 volt section output voltage reaches 10±0.25 volts for a duration of 20±1 milliseconds. The 11 volt section shall meet the requirements of 3.2.1.2.1.2 and 3.2.1.3.1. The ±30 volt section shall meet the requirements of 3.2.1.2.2.3, 3.2.1.3.2, and 3.2.1.3.3."

* 4.5.2.9.2: Delete the following sentence:

"The battery shall meet the requirements of 3.2.1.1.1, 3.2.1.1.1.1, 3.2.1.1.2.1, 3.2.1.1.3, and 3.2.1.1.4."

and substitute the following:

"The battery shall meet the requirements of 3.2.1.1.1.1, 3.2.1.1.2, 3.2.1.1.2.1, 3.2.1.1.3 and 3.2.1.1.4. The battery shall meet the requirements of 3.2.1.1.1 with the exception that one battery shall be allowed to have a voltage as low as 8 volts between 70 and 100 seconds."

4.5.3.1: Delete the second sentence and substitute the following:

"The operating temperature shall be 145° ± 2.5°F (62.7 ± 1.4°C), and the temperature soak prior to battery activation shall be 4 hours minimum."

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4.5.3.2: Delete the second sentence and substitute the following:

"The operating temperature of Step 4 shall be -25°F ± 2.5°F (-31° ± 1.4°C) and the temperature soak prior to battery activation shall be 4 hours minimum."

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4.5.3.5: Delete entire paragraph and substitute the following:

"4.5.3.5 Transportation vibration test.- Affix the thermal battery to a vibration table in a manner which will simulate its normal mounting. Subject the thermal battery to the vibration tests of MIL-STD-810. Method 514.2, Procedure X using the criteria of Table 514.2-VII and Figure 514.2-7 curve AX, except the dwell shall be as follows, 35 Gs double amplitude at 110 Hz for 20 seconds in each of 3 axes. Following exposure to this environment the thermal battery shall meet the requirements of 3.2.1.2.1.1, 3.2.1.2.2.1, 3.2.1.2.2.2, 3.2.1.6, and 3.2.1.7."

4.5.3.8: Delete the first sentence and substitute the following:

"The thermal battery shall be installed in an appropriate canister in a manner that will facilitate activation of the 11 V section and the assembly exposed to a launch shock of 900 to 1100 G, from an 8 inch gun, M115 or equivalent (GFE), at 70°F ± 30°F (21.1°C ± 16.7°C)."

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* 4.5.5, line 3, delete "9301200" and substitute with "9304695".

PAGE 15

* Delete Figure 1 entirely and substitute the attached Figure 1.

PAGE 16

* Delete Figure 2 entirely and substitute the attached Figure 2.

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* Delete Figure 3 entirely and substitute the attached Figure 3.

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* Delete Figure 4 entirely and substitute the attached Figure 4.

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* Delete Figure 5 entirely and substitute the attached Figure 5.

PAGE 20

* Delete Figure 6 entirely and substitute the attached Figure 6.

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* Delete Figure 7 entirely and substitute the attached Figure 7.

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* Delete Table I entirely and substitute the attached Table I.

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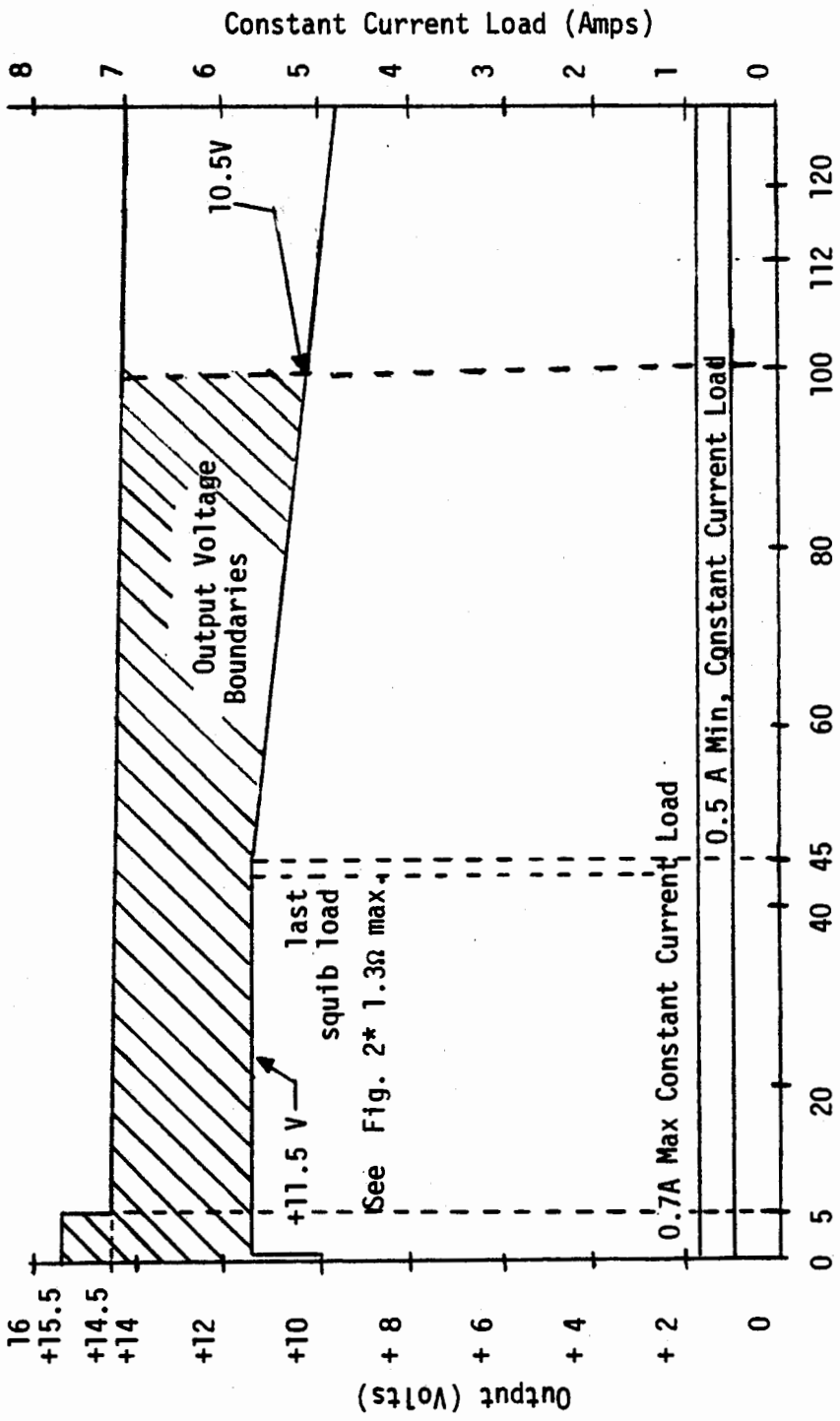


Figure 1. +11 Volt Power Source Voltage and Load Profiles (0 to 100 Seconds)

* Refer to paragraph 3.2.1.1.1.1 for times at which squib loads can occur, minimum voltage permissible during squib loads, and recovery time.

NOTE: Combined load consists of constant current and resistive loads.

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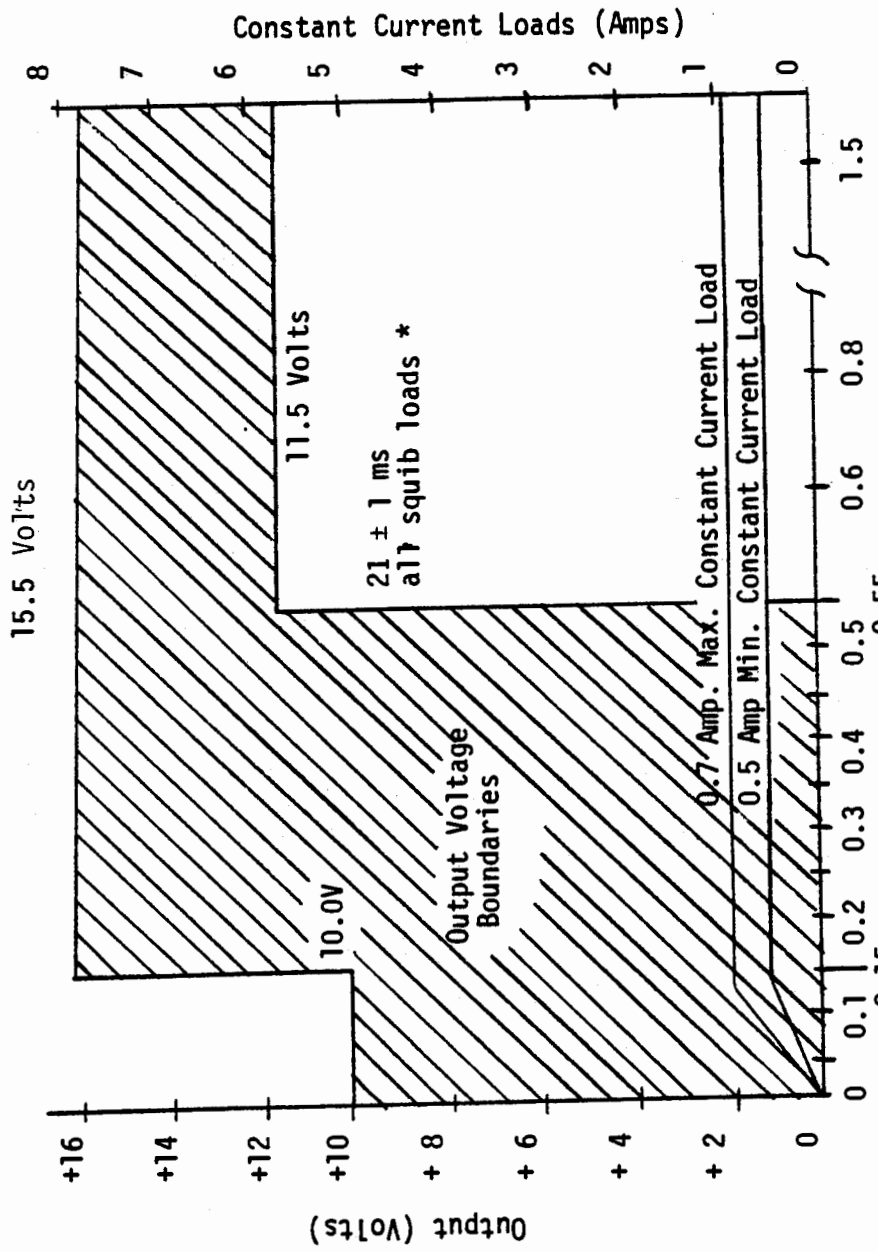


Figure 2. +11 V Power Source Voltage and Load Profiles (0 to 1.5 seconds)

* Refer to paragraph 3.2.1.1.1.1 for times at which squib loads can occur, minimum voltage permissible during squib loads, and recovery time.

NOTE: Combined load consists of constant current and resistive loads.

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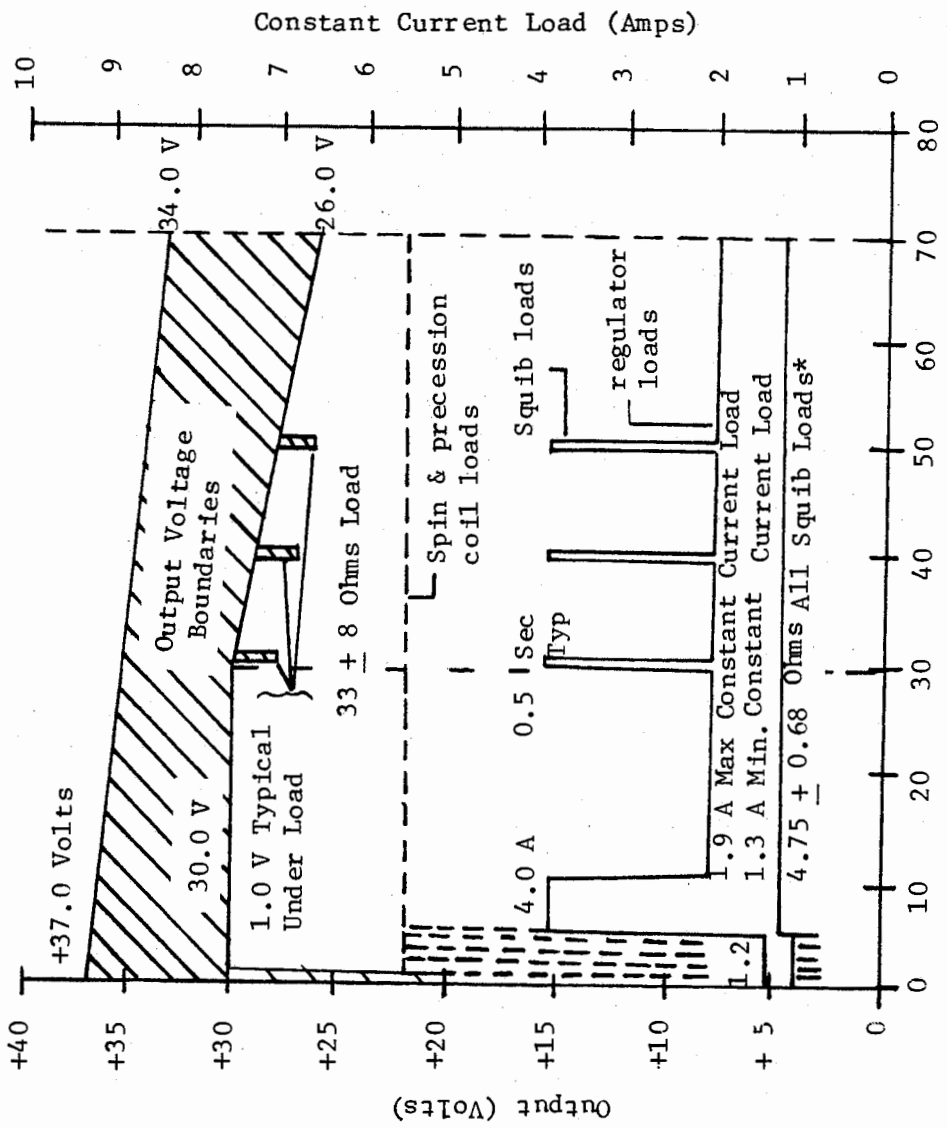


Figure 3. +30 Volt Power Source Voltage and Load Profiles (0 to 70 Seconds)

* Refer to paragraph 3.2.1.1.2.1 for times at which squib loads can occur, minimum voltage permissible during squib loads, and recovery time.
NOTE: Combined load consists of constant current and resistive loads.

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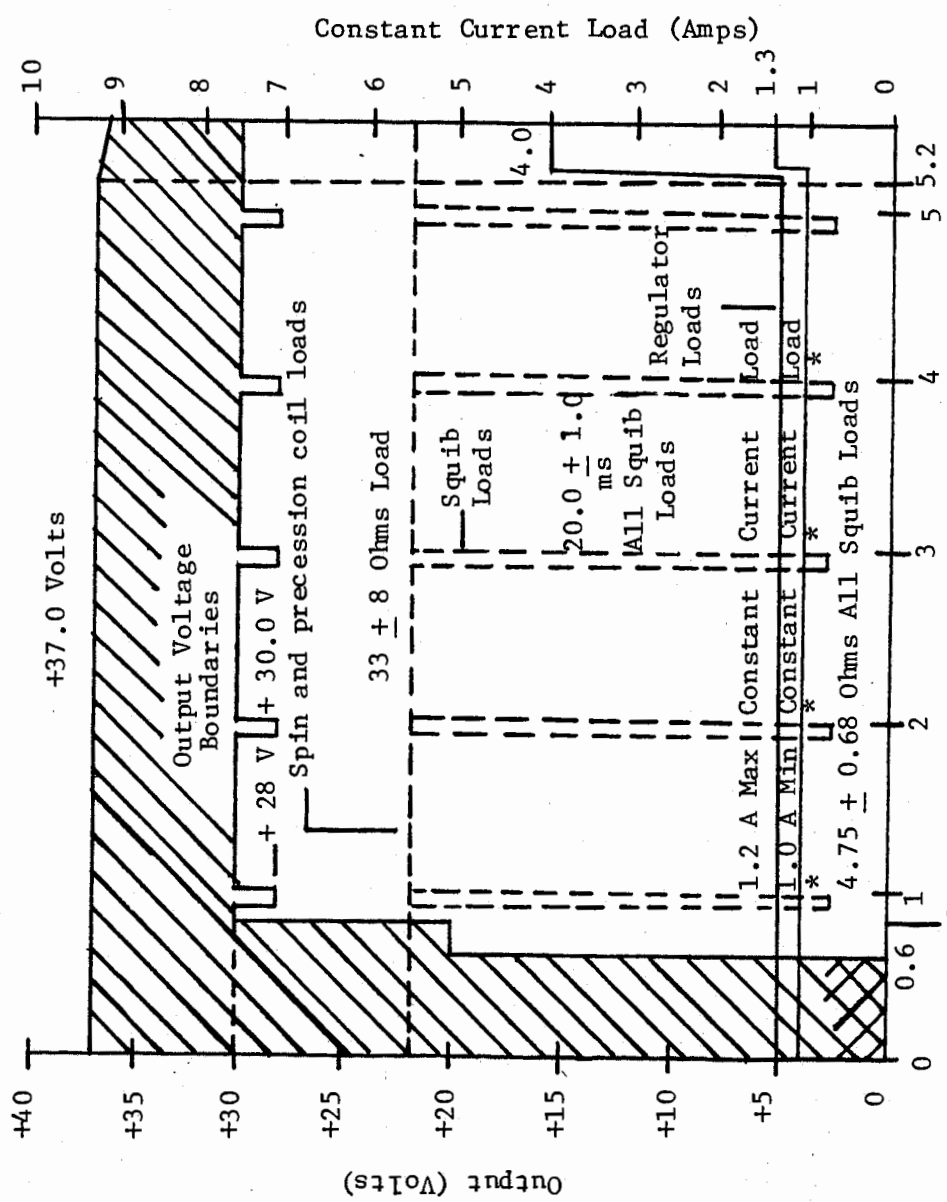


Figure 4. +30 Volt Power Source Voltage and Load Profiles (0 to 5.5 Seconds)
(Typical Squib Load Pattern Shown)

* Refer to paragraph 3.2.1.1.2.1 for times at which squib loads can occur, minimum voltage permissible during squib loads and recovery time.

NOTE: Combined load consists of constant current and resistive loads.

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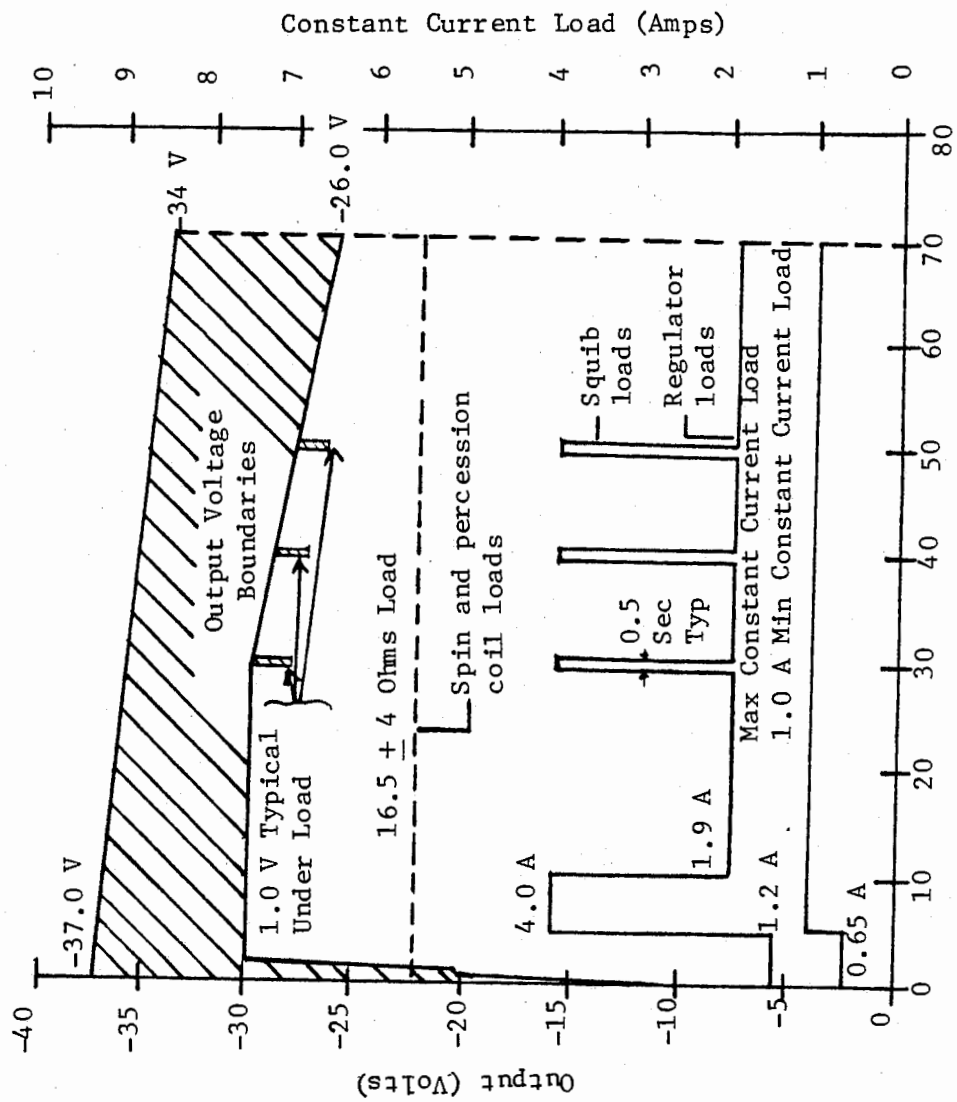


Figure 5. -30 Volt Power Source Voltage and Load Profiles (0 to 70 Seconds)

NOTE: Combined load consists of constant current and resistive loads.

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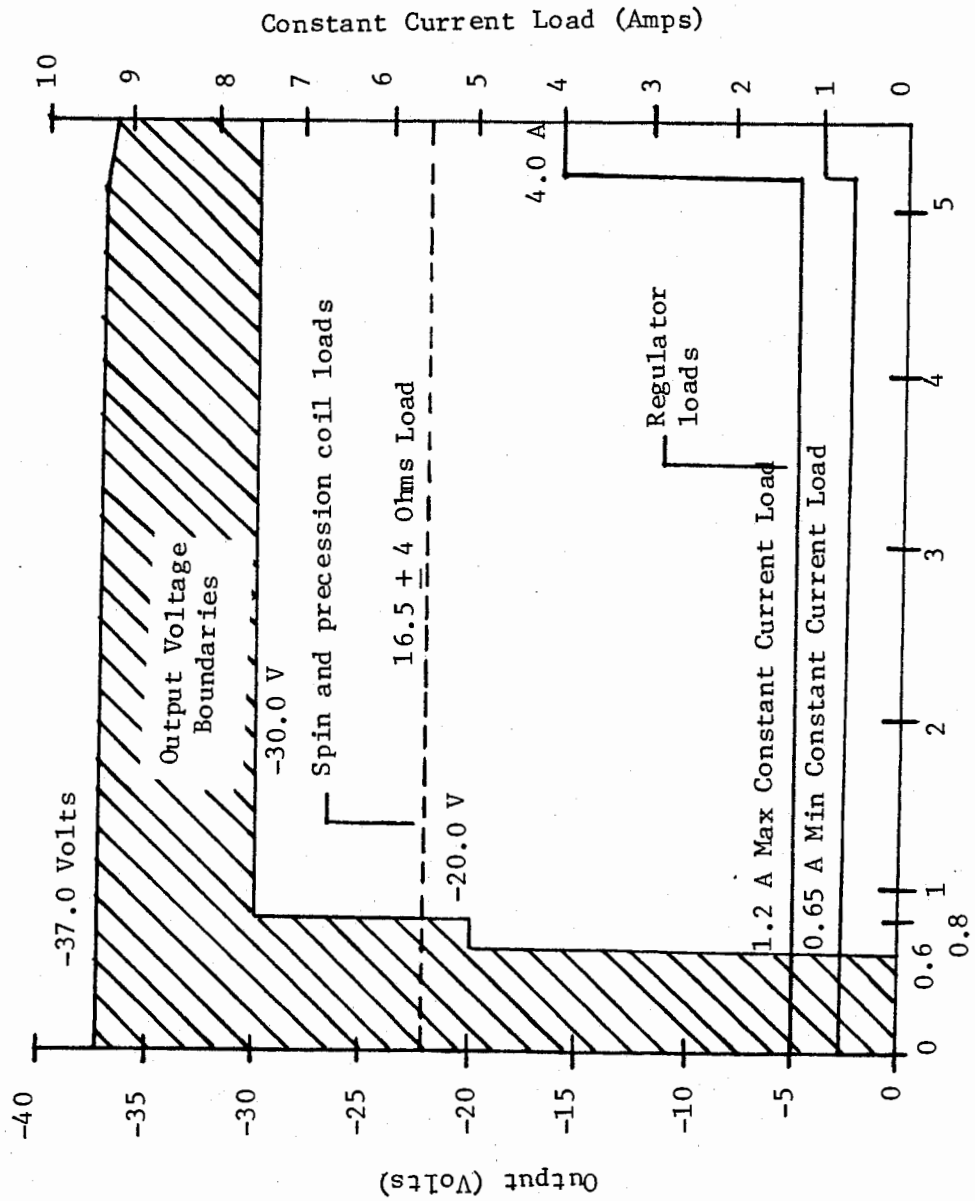


Figure 6. -30 Volt Power Source Voltage and Load Profiles (0 to 70 Seconds)

NOTE: Combined load consists of constant current and resistive loads

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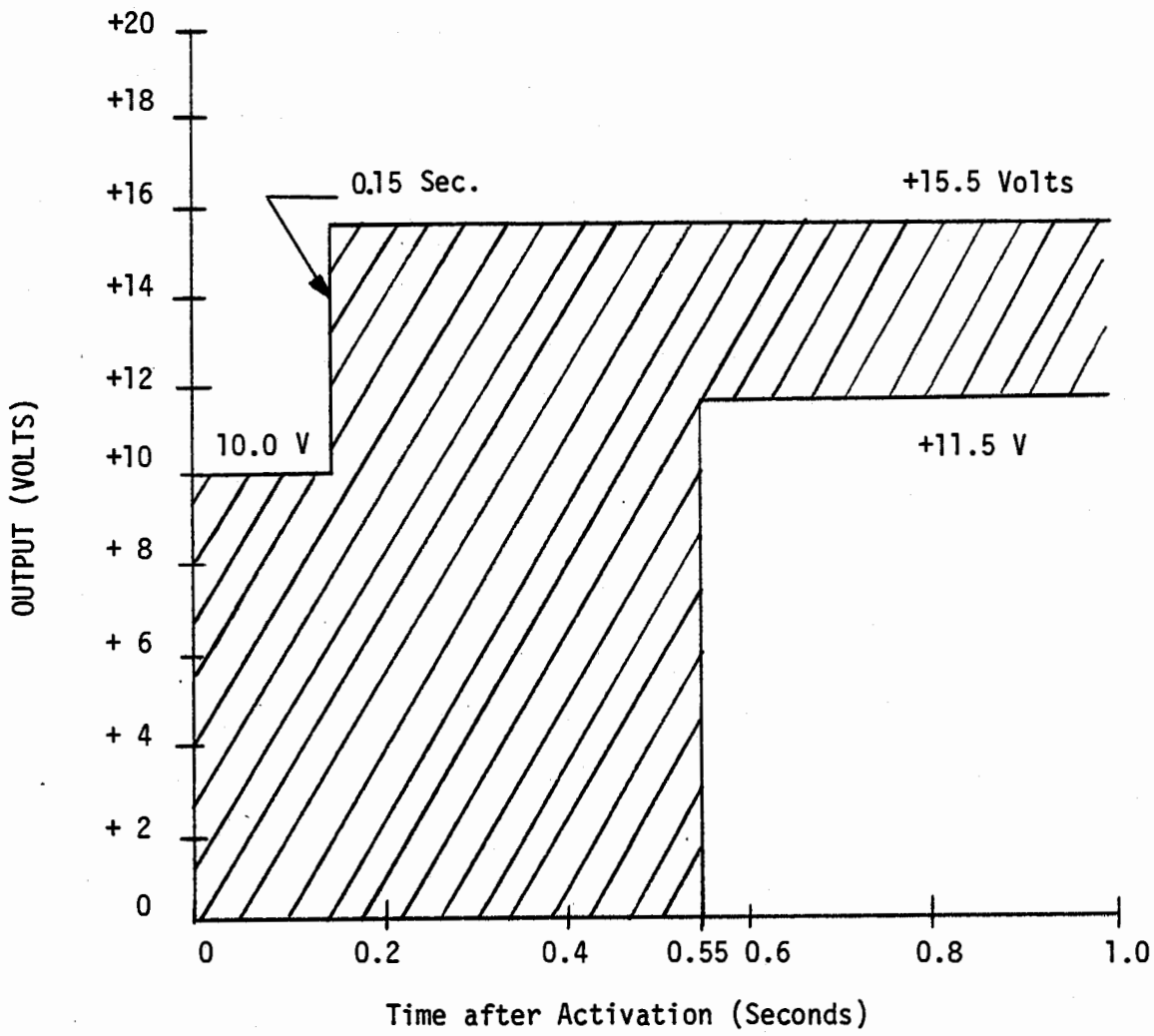


Figure 7. Output Voltage Rise Time Boundaries for +11 Volt Source

Note: Refer to paragraph 3.2.1.1.1.1 for times at which squib loads can occur, minimum voltage permissible during squib loads, and recovery time.

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TABLE I

+11 VOLT SOURCE APPLIED SQUIB LOADS

SQUIB LOAD	LOAD (NOTE 1)	APPLICATION TIME	MINIMUM VOLTAGE	MAXIMUM RECOVERY TIME
	OHMS		VOLTS	
1	31.0 max	When output voltage reaches 10.0 volts min.	7.5	Note 2
2	2.5(145°F) max. 1.9(-25°F) max.	.210 seconds after first squib is applied	Note 3	Note 2
3	1.3 max.	1.5 seconds after first squib is applied	8.0	75 ms
4	1.3 max.	45.0 seconds after first squib is applied	8.0	75 ms

NOTE 1. Squib loads applied for 20.1 $\pm$ 0.1 milliseconds.

NOTE 2. Voltage recovery time is 100 milliseconds minus squib load application time.

NOTE 3. Battery voltage shall fire the $\pm$ 30 volts squib and shall not fall below 8.6 volts (at -25°F) or 11.3 volts (at +145°F).

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* Table II, Sheet 1: Delete Drawing Number "9301200" and substitute with "9304695".

* Table II, Sheet 1: Delete major defects 104 and 105 and substitute with the following:

"104 ±30V no fire current (Note 4) 100% 3.2.1.2.2.2 4.5.2.7"

"105 +11V no fire stimulus (Note 5) 100% 3.2.1.2.1.1 4.5.2.8"

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* Table II, Sheet 2: Delete Drawing Number "9301200" and substitute with "9304695".

* Table II, Sheet 2: Add the following notes:

"4) Those units tested for items 106 thru 110 shall be tested once for item 104 after tests of items 106 thru 109."

5) Those units tested for items 106 thru 110 shall be tested once for item 105 before test of items 106 thru 109."

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* Table III, Sheet 1: Delete Drawing Number "9301200" and substitute with "9304695".

Table III, Sheet 1: Delete major defect 110 entirely and substitute the following:

"110±30V bridge resistance 100% 3.2.1.2.2.1 4.5.2.6."

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* Table III, Sheet 2: Delete Drawing Number "9301200" and substitute with "9304695".

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* Table IV, Sheet 1: Delete Drawing Number "9301200" and substitute with "9304695".

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* Table IV, Sheet 1: Delete major defects 103 and 104 and substitute with the following:

"103 ±30V no fire current (Note 2) 1.0% 3.2.1.2.2.2 4.5.2.7"

"104 +11V no fire stimulus (Note 3) 1.0% 3.2.1.2.1.1 4.5.2.8"

* Table IV, Sheet 1: Add the following notes:

"2) Perform test for item 103 once after testing for item 105."

"3) Perform test for item 104 once before testing for item 105."

NOTE: The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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